Rossmann Store Sales Forecasting Using Machine Learning

I. Dataset Description

The project will use the Rossmann Store Sales dataset, which contains historical daily sales records from 1,115 Rossmann drugstores in Germany. The uploaded training dataset consists of 1,017,209 records covering January 1, 2013, to July 31, 2015, with nine columns: Store, DayOfWeek, Date, Sales, Customers, Open, Promo, StateHoliday, and SchoolHoliday.

The dataset provides information about historical store sales, customer counts, store operations, promotions, and holidays. The target variable is Sales, which represents the daily total sales amount of each store rather than individual product sales. The dataset will be used to identify historical patterns and develop a model for forecasting daily store sales up to 42 days (six weeks) ahead.

Dataset Source: Kaggle – Rossmann Store Sales
https://www.kaggle.com/competitions/rossmann-store-sales/data

License: The dataset page does not specify a separate open-source license. The dataset was really intended for the competition conducted by kaggle. We only used this dataset for academic purposes only, we don't have a plan to join the competition.

II. Problem Statement

Inaccurate sales forecasting can cause operational problems for retail businesses, particularly overstocking, understocking, and staffing mismatches. Overstocking may result in unnecessary inventory costs, while understocking can lead to product shortages and missed sales opportunities. Similarly, inaccurate sales estimates can result in insufficient or excessive staffing.

This project aims to develop a machine learning-based forecasting model that predicts the daily sales of each Rossmann store for the next six weeks. By using historical sales, promotional activities, weekdays, and holiday information, the model is expected to provide more reliable sales estimates that can support inventory planning and staffing decisions.

III. Chosen Metric

The project will use Mean Absolute Error (MAE) as the primary evaluation metric, supported by Root Mean Squared Error (RMSE) and Weighted Absolute Percentage Error (WAPE).

- **MAE** will measure the average absolute difference between actual and predicted sales.
- **RMSE** will give greater weight to larger prediction errors.
- **WAPE** will measure the overall forecasting error relative to total actual sales.

MAE will serve as the main basis for comparing model performance, while RMSE and WAPE will provide additional insights into forecasting accuracy. Lower values indicate better forecasting performance. These metrics will help determine whether the model produces reliable predictions for store-level sales planning.

IV. Baseline Idea

A Seasonal Naive Forecasting method will serve as the baseline. It will use the most recent available sales for each weekday and repeat this seven-day sales pattern across the 42-day forecasting period. This approach will provide a simple reference point for determining whether the selected machine learning algorithms offer meaningful improvements.